STRATHMORE UNIVERSITY

SCHOOL OF COMPUTING AND ENGINEERING SCIENCES

MACHINE LEARNING-BASED TOOL FOR EARLY DETECTION OF AT-RISK STUDENTS IN HIGHER EDUCATION: CASE OF STRATHMORE UNIVERSITY

BY

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DECLARATION

Student's Declaration

I declare that this thesis is my original work and has not been presented to any other institution for examination or any other award. Where other sources of information have been used, they have been properly acknowledged.

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Supervisor's Declaration

This thesis has been submitted for examination with my approval as the University Supervisor.

Name: Dr. [Supervisor Name]

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Contents

Table of Abbreviations

Abbreviation	Description
AMS	Academic Management System
CBET	Competency-Based Education and Training
CESA	Continental Education Strategy for Africa (2016-2025)
EDM	Educational Data Mining
GPA	Grade Point Average
LMS	Learning Management System
ML	Machine Learning
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
SHAP	SHapley Additive exPlanations
SIM	Student Integration Model
SVM	Support Vector Machines

Chapter 1

INTRODUCTION

1.1 Background to the Study

Student retention and academic success represent critical challenges confronting higher education institutions worldwide. According to (Irwin et al., 2024), undergraduate enrollment in degree-granting institutions decreased by 13% between 2012 and 2022, while 23.3% of undergraduate students leave universities annually. Six-year graduation rates reveal substantial variation: 63% at public universities, 68% at private nonprofit institutions, and only 29% at private for-profit institutions. This persistent attrition represents significant personal and economic costs for individual students, alongside substantial challenges for institutional sustainability and mission fulfillment.

Technology is changing that. Machine learning algorithms can now identify at-risk students weeks or even months before traditional warning signs appear, achieving prediction accuracies of 88–92% (Ahmed, 2024; Gathani et al., 2026). Georgia State University's GPS Advising system illustrates what becomes possible: by tracking over 800 risk factors for 40,000 students daily and generating 90,000 targeted interventions each year, the university raised its graduation rate by 7.8 percentage points and virtually eliminated achievement gaps between demographic groups (Georgia State University, 2024). The technology, in short, is mature and proven—but its benefits have not yet reached most of the world.

Africa is a striking example of that gap. The African Union's Continental Education Strategy (CESA 2016–2025) acknowledges retention as a priority, yet most African universities still rely on end-of-semester transcripts as their primary diagnostic tool. Some

sub-regions have begun to experiment. Mohammed V University in Morocco achieved an 81.66% dropout-prediction accuracy in a pilot study that utilized L2-regularized logistic regression (Bouihi et al., 2024). "Similarly, a study at the University of Johannesburg in South Africa utilized machine learning to predict the performance of 1,366 engineering students, achieving a 65.86% accuracy rate using Decision Tree algorithms and identifying prior performance in physics and mathematics as primary success factors (Taodzera, 2016).

Within East Africa, Kenya's higher education sector faces particularly acute retention pressures. Government capitation covers only 57% of students against an 80% target, while public universities have accumulated Ksh 62 billion in unpaid bills—structural constraints that make student attrition both more likely and more financially consequential for students, families, and institutions (Daily Nation, 2024; Kenya Institute for Public Policy Research and Analysis, 2024).

Recognizing the challenge of student attrition, researchers from the University of Nairobi in collaboration with one researcher at the Free University of Brussels conducted a study utilizing Machakos University College as a case study to apply Business Intelligence and machine learning to dropout prediction (Ngemu et al., 2015). Using the WEKA toolkit to analyze historical institutional data, the research team—comprising Ngemu, Odongo, Omulo, and Manderick—evaluated multiple classification algorithms, including Naïve Bayes, Support Vector Machines (SVM), and the J48 decision tree. Their findings identified that students at-risk of dropping out could be accurately flagged based on critical predictors such as financial stability (fee payment), entry age, health status, and parental occupation (Ngemu et al., 2015).

Strathmore University operates within this challenging global landscape. While maintaining distinct academic standards that intensify the consequences of delayed intervention. The institution enforces a 67% minimum attendance threshold for examination eligibility and a 2.0 GPA floor for good academic standing—policy thresholds that render early identification especially critical, as students who breach them may be barred

from examinations or placed on academic probation regardless of their potential for recovery. The university collects extensive student data through its Academic Management System (AMS), Learning Management System (LMS), and attendance tracking platform, generating daily records on academic performance, engagement behaviour, and class participation.

Despite this data richness, these systems operate as isolated administrative repositories with no integrated analytics capability, preventing the transformation of raw institutional data into actionable early warnings. Students are typically identified for intervention only after manifesting metrics confirming trends and outcomes after events have occured. This reactive paradigm, though common across Kenyan institutions, is inconsistent both with international best practice (Georgia State University, 2024) and with Kenya's national call for greater technology utilisation in higher education (Republic of Kenya Ministry of Education, 2024). This study addresses that gap directly: by developing a machine learning-based predictive analytics platform calibrated to Strathmore's academic policies, data infrastructure, and resource constraints, it seeks to transform the institution's support model from reactive monitoring to proactive, evidence-driven intervention.

1.2 Statement of the Problem

Strathmore University currently faces a critical gap between data collection and actionable student support. Although the institution maintains robust data streams through its Academic Management System (AMS), Learning Management System (LMS), and attendance tracking platforms, these systems operate as isolated administrative repositories rather than integrated predictive tools. This fragmentation prevents the university from transforming raw institutional data into early warnings for at-risk undergraduate students.

The primary consequence of this fragmentation is a reactive support paradigm that carries measurable human costs. Students are typically identified for intervention only

after manifesting indicators of failure—cumulative low grades, breach of the mandatory 67% attendance threshold for examination eligibility, or formal withdrawal notice. By this stage, the window for effective academic recovery has often closed, and the outcomes that follow are both predictable and preventable: students drop out of programmes they could have completed, fail courses that timely tutoring referrals could have salvaged, and accumulate delays that extend their studies by entire semesters—each outcome representing a personal and institutional loss that earlier detection would have averted.

Despite possessing the necessary digital infrastructure and longitudinal student datasets, Strathmore University lacks a specialised machine learning architecture capable of integrating these data streams into a coherent risk stratification system. This absence is consequential: without the capability to detect deteriorating engagement trajectories weeks before they manifest as failed grades or breached attendance thresholds, the institution cannot intervene at the point when intervention is still effective. Without an automated ML-based tool to predict dropout risk, course failure risk, and program delay risk simultaneously, the university remains unable to allocate its limited counselling and mentoring resources to the students who need them most—and advisors are left to act on intuition rather than evidence, on crisis rather than prevention.

This study therefore develops, implements, and evaluates an integrated predictive analytics ML-based tool tailored to Strathmore University's academic policies, data infrastructure, and resource constraints. By producing explainable and actionable risk predictions across three distinct outcomes, the ML-based tool is designed to transition the institution from reactive to proactive monitoring and data-driven intervention.

1.3 Research Objectives

1.3.1 Main Objective

To design, develop, and evaluate a machine learning-based predictive analytics tool for early identification of at-risk undergraduate students at Strathmore University.

1.3.2 Specific Objectives

1. To design a Machine Learning predictive tool for university administrators, school administrators, and academic mentors that present actionable risk predictions with supporting intervention guidance.
2. To evaluate model accuracy and predictive performance using historical student cohort data (2021-2024) and standard machine learning evaluation metrics including Accuracy, Precision, Recall, F1-score, and ROC-AUC.
3. To assess the platform's usability and perceived institutional impact through structured evaluation with academic advisors, school administrators, and mentors, with specific attention to intervention feasibility and long-term sustainability.

1.4 Research Questions

1. How can risk predictions be effectively presented to different stakeholder groups—university administrators, school administrators, and academic mentors—to support informed decision-making?
2. To what extent does training predictive models on historical student cohort data (2021-2024) with known outcomes produce valid and generalisable predictions for current enrolled students?
3. How do academic advisors, school administrators, and mentors perceive the platform's usability, intervention feasibility, and potential for long-term institutional

adoption?

1.5 Justification

1.5.1 Operational Necessity

This research is justified by the need to bridge the gap between data collection and data utilization in Universities such as Strathmore University. Currently, Strathmore's student support is reactive—interventions. It often occurs only after a student has already failed or dropped out. There is an urgent need for a system that transforms underutilized data into actionable early warnings without increasing the administrative burden on faculty.

1.5.2 Addressing the "Black Box" Problem

This study is justified by its focus on explainability. By using specialized techniques to show why a student is at risk, the research justifies the shift toward evidence-based mentoring, ensuring that institutional support is credible.

1.5.3 Policy and Economic Pressures

This study responds to the Kenya Ministry of Education's Sessional Paper No. 1 of 2024, which cites inadequate technology use as a core bottleneck in higher education (Kenya Institute for Public Policy Research and Analysis, 2024). In a climate of rising enrollments and limited government funding, the status quo of universal mentoring is no longer sustainable. This research is justified as an economic necessity to ensure that finite counseling resources are directed only where they are most needed.

1.6 Scope and Delimitations of the Study

1.6.1 Scope

This research focuses on the development and evaluation of a predictive analytics platform for approximately 5,000 undergraduate students across all five schools at Strathmore University, Kenya. The technical scope involves integrating data from the Academic Management System (AMS), Learning Management System (LMS), and attendance tracking records into a unified analytics pipeline. The study develops three prediction models—targeting dropout risk, course failure risk, and program delay risk—using ensemble algorithms including Random Forest, Gradient Boosting, and XGBoost, trained on historical student cohort data from 2021-2024 with known academic outcomes. Model predictions are accompanied by SHAP-based explainability outputs to ensure advisors and administrators can understand and act on the reasons behind each risk classification (Lundberg & Lee, 2017). The resulting platform includes role-based dashboards for university administrators, school administrators, and academic mentors, with model performance measured through standard machine learning evaluation metrics (Accuracy, Precision, Recall, F1-score, and ROC-AUC) and platform effectiveness assessed through structured stakeholder feedback.

1.6.2 Delimitations

The research boundaries are defined by the following deliberate constraints. Regarding data factors, the analysis is restricted to academic risk factors measurable through institutional systems, such as attendance rates, GPA, LMS engagement, and course grades. External variables such as socioeconomic background, mental health status, and family circumstances are excluded, as they are not systematically captured in Strathmore's existing data infrastructure and their collection would raise ethical concerns beyond the current scope. In terms of student population, the study is limited to undergraduate students across Strathmore University's five schools; postgraduate and continuing

education cohorts are excluded due to their unique assessment patterns.

The system capabilities are delimited to identifying risk levels and suggesting general intervention categories, such as academic counselling or tutoring referrals, rather than providing course-specific remediation. Within the algorithmic scope, the study employs ensemble tree-based methods like Random Forest and XGBoost (Chen & Guestrin, 2016) rather than deep learning architectures. This decision is justified by the dataset size of approximately 5,000 students and the critical need for feature interpretability. Furthermore, the technical platform is developed as a web-based application, with mobile-native development remaining outside the scope. Financial domains, including fee payment risk and scholarship management, are also excluded to focus purely on academic and engagement indicators. Finally, the implementation phase concludes with a functional prototype and structured evaluation; full-scale production deployment and long-term longitudinal impact assessment are reserved for future research.

1.7 Definition of Terms

Term	**Description**
At-Risk Student	A student whose data patterns statistically indicate a high probability of dropout, course failure, or program delay prior to the actual event.
Dropout Risk	The predicted probability of a student discontinuing enrolment before completing their degree requirements.
Course Failure Risk	The predicted likelihood of a student failing a course or becoming exam-ineligible due to attendance falling below the mandatory 67% threshold at Strathmore University.
Program Delay Risk	The probability that a student will exceed the standard programme duration due to academic probation, course repetition, or accumulated failed semesters.

Term	Description
Academic Management System	The primary institutional database for official student records, including demographic information, enrolment status, and academic history.
Learning Management System	A digital platform used to deliver course content and track student engagement through behavioural data such as login frequency, resource access, and assignment submissions.
Early Warning System	An integrated tool that uses data and predictive modelling to trigger timely alerts for students identified as requiring academic intervention.
Engagement Metrics	Quantitative measures of student participation derived from institutional systems, such as LMS login frequency, assignment submission timeliness, and attendance rate trajectories.
Ensemble Learning	A machine learning technique that combines multiple algorithms—such as Random Forest, Gradient Boosting, and XGBoost—to produce predictions that are more accurate and stable than any single model alone.
Feature Engineering	The process of transforming raw institutional data into optimised input variables for machine learning model training, including the creation of derived indicators such as attendance flags and GPA thresholds.
Intervention	A proactive support action—such as academic mentoring, tutoring referral, or advisor outreach—triggered by a student's identified risk status and designed to prevent academic failure or withdrawal.
Predictive Analytics	A branch of data analytics that uses statistical modelling and machine learning techniques applied to historical data to forecast future outcomes.

Term	Description
Explainability	The capacity of a predictive model to provide human-interpretable reasons for its predictions, often implemented through SHAP values.
Historical Cohort	A defined group of students whose complete academic trajectories—from enrolment to graduation—are known and documented.
Model Accuracy	The proportion of correct predictions (both at-risk and not-at-risk) made by a model relative to total observations.
Precision	The proportion of students predicted as at-risk who are genuinely at-risk. High precision minimises false alarms.
Recall (Sensitivity)	The proportion of genuinely at-risk students who are correctly identified by the model.
F1-Score	The harmonic mean of Precision and Recall, providing a single balanced metric.
ROC-AUC	A threshold-independent metric measuring a model's ability to discriminate between classes. A value of 1.0 indicates perfect discrimination.

1.8 Significance

1.8.1 Theoretical and Methodological Significance

This study advances the field of Educational Data Mining (EDM) by operationalizing Tinto's Student Integration Model within an African institutional context. It provides empirical evidence that Western retention theories can be mapped to quantifiable data elements. Methodologically, it demonstrates that SHAP-based explainability is not a luxury but a prerequisite for AI adoption in resource-constrained environments, setting a new standard for how predictive models should be evaluated beyond simple accuracy.

1.8.2 Practical Significance

For Institutions: The study provides a documented, open-source blueprint for transitioning from reactive to proactive support. It offers a sustainable alternative to the individual-dependency models that have failed in other regional universities by embedding automated data pipelines into the university's core infrastructure.

For Educators: The platform empowers advisors with role-based dashboards that move decision-making from intuition to evidence. It transforms the advisor's role from a monitor to a facilitator, allowing for substantive, reason-led conversations based on specific risk drivers.

For Students: As the primary beneficiaries, students gain a safety net. The system transforms punitive policies (like the 67% attendance rule) into supportive frameworks, allowing for academic recovery before thresholds are irreversibly breached.

1.8.3 National and Policy Significance

At the national level, this research serves as a replicable model for the Republic of Kenya's transition to Competency-Based Education and Training (CBET). By demonstrating how a mid-sized university can implement high-level analytics within tight budget constraints, the study provides a policy instrument for other East African institutions to improve retention rates without the prohibitive costs of commercial software.

1.9 Summary

This chapter established the context, rationale, and direction of the study. Beginning at the global level, it demonstrated that student dropout remains a critical challenge in higher education and that machine learning-based predictive analytics has proven capable of identifying at-risk students weeks before conventional indicators surface—yet these benefits remain largely inaccessible to number of African universities.

The logic of disparate systems forces institutions like Strathmore into a reactive sup-

port paradigm where students are identified only after breaching the 67% attendance threshold or 2.0 GPA floor. This gap, situated within Kenya's national call for greater technology utilization in education, defines the problem this study addresses.

The study responds by developing, ML-based predictive analytics platform tailored to Strathmore's policies and resource context. The platform predicts dropout, course failure, and programme delay using ensemble algorithms trained on the 2021–2024 historical cohort. Its contributions span theory, practice, and policy: advancing retention frameworks in African contexts, providing a sustainable and replicable institutional tool, and demonstrating a cost-effective model for comparable universities across the region. Chapter Two now reviews the literature underpinning these contributions, examining theoretical frameworks, documented system implementations with their successes and limitations, and the precise knowledge gaps this study is designed to fill.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

The purpose of this chapter is to synthesize the multidisciplinary body of research that underpins this study's design and motivation. The review spans three interconnected domains: the theoretical frameworks governing student retention, the global evolution of predictive analytics and Educational Data Mining (EDM) in higher education, and the African and Kenyan institutional contexts in which this research is embedded. Together, these streams establish both the intellectual foundation and the practical necessity of a machine learning-based Early Warning System (EWS) calibrated to Strathmore University's academic policies and data infrastructure.

The chapter is organized to trace a deliberate logical arc. First, it begins with the macroeconomic and human costs of student attrition. Second, it narrows progressively from global best practices to regional pilot studies. And finally, it focuses on the specific institutional gaps this research is designed to fill. Central to this arc is the argument that modern machine learning does not merely add technological convenience; it operationalizes classical retention theory, transforming abstract constructs such as academic integration and social engagement into measurable and actionable data signals.

2.2 Global Perspectives on Student Attrition

Student retention is a global imperative linked directly to both institutional sustainability and the fulfillment of higher education's social mission. According to Irwin et al. (2024), undergraduate enrollment in degree-granting institutions in the United States

decreased by 13% between 2012 and 2022, compounded by a persistent annual attrition rate of 23.3%. Six-year graduation rates reveal substantial structural disparities: 68% at private nonprofit institutions, 63% at public universities, and only 29% at private for-profit institutions. Despite a modest recent recovery—the 2024 National Student Clearinghouse report recorded a national first-year retention rate of 68%, described as the highest in the past decade (National Student Clearinghouse Research Center, 2024)—the structural problem of attrition remains unresolved, with large proportions of students continuing to exit before degree completion.

These figures carry dual consequences. At the institutional level, each departing student represents lost tuition revenue, wasted administrative investment, and diminished institutional capacity to fulfill its educational mission. At the individual level, students who withdraw often exit the system bearing financial debt without the credential necessary for socioeconomic mobility. A systematic review of 125 peer-reviewed studies on undergraduate retention published between 2014 and 2022 concluded that, despite decades of research and institutional intervention, student dropout remains a persistent and structurally significant problem across higher education globally, with research in the field remaining fragmented and without a unified evidence base for effective remediation (Gonçalves et al., 2024). The following table summarizes graduation rate variation by institution type:

Institution Type	**Six-Year Graduation Rate**
Private Nonprofit Institutions	68%
Public Universities	63%
Private For-Profit Institutions	29%

Table 2.1: Six-Year Graduation Rates by Institution Type (Irwin et al., 2024)

Beyond the United States, the problem is not geographically bounded. Widening access policies across the Global South have expanded enrollment without proportional improvements in completion, meaning that participation gains mask a deepening com-

pletion deficit (Gonçalves et al., 2024). In sub-Saharan Africa and Kenya specifically, this deficit is compounded by resource constraints, infrastructure gaps, and the financial pressures documented in Section 2.7, making the attrition problem both more acute and less resourced than in high-income institutional contexts. These global statistics therefore establish the urgency of the problem and demonstrate that student attrition is not a peripheral concern but a structural threat to the viability of higher education institutions worldwide—providing the macro-level context within which Kenya's institutional challenges, and this study's intervention, must be understood.

2.3 Theoretical Frameworks for Student Retention

2.3.1 Tinto's Student Integration Model

The dominant theoretical anchor for student retention research remains Vincent Tinto's Student Integration Model (SIM). First proposed by Tinto and Cullen in 1973 and formally published as a peer-reviewed synthesis in 1975, the model draws its conceptual foundations from Émile Durkheim's theory of suicide— specifically the proposition that a deficit of social cohesion precipitates withdrawal from the community (Tinto, 1975). Tinto's scholarly influence has been exceptional: his works have been cited in more than 700 studies across multiple disciplines, establishing the SIM as the most widely adopted theoretical lens in higher education retention research (Bento Júnior et al., 2025).

At its core, the SIM conceptualizes the university as a microcosm comprising two interdependent systems—the academic and the social—and posits that a student's persistence is governed by their degree of integration into both (Tinto, 1975, 1993). Critically, Tinto frames dropout not as a single discrete event but as a *longitudinal process*: the student's commitment to persist is continuously reconfigured through ongoing interactions. This interactional character means that dropout cannot be attributed to any one variable in isolation; it emerges from the cumulative weight of how individual at-

tributes, academic performance, and social relationships interact and accumulate over time (Nicoletti, 2019; Tinto, 1993).

2.3.1.1 The Longitudinal Mechanism

The SIM operates through a structured, multi-stage sequence linking pre-institutional characteristics to eventual persistence:

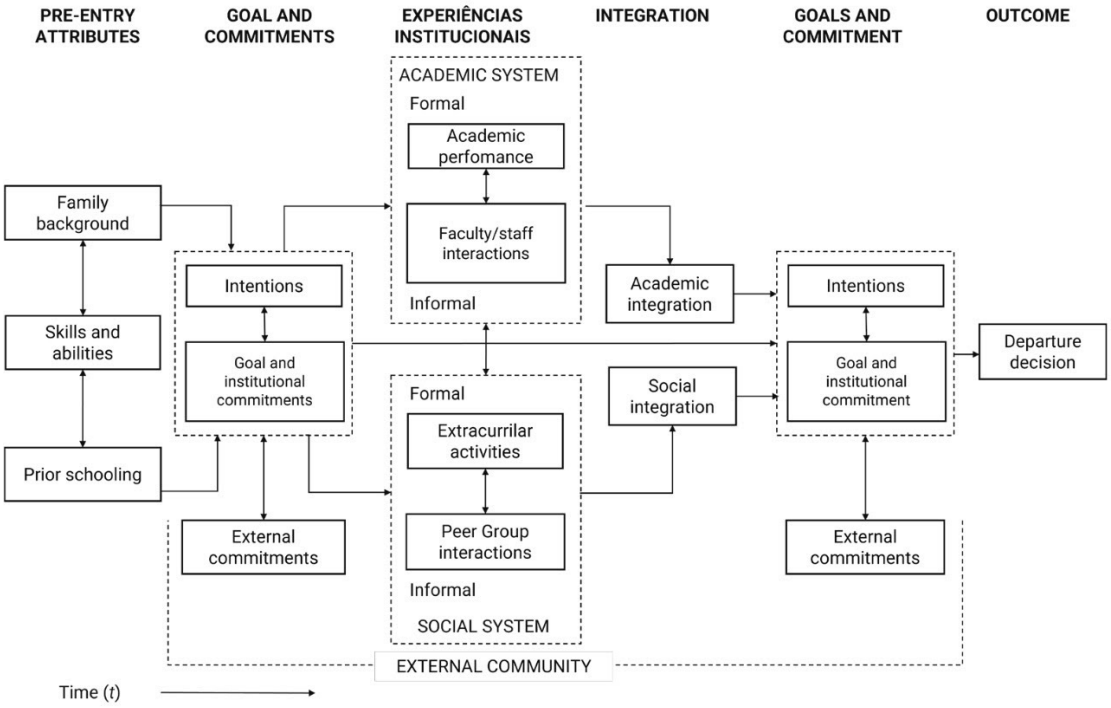

Figure 2.1: Tinto's Model of Institutional Departure

1. **Pre-Entry Attributes:** Students enter the institution carrying three foundational background variables: family background, individual skills and abilities, and prior schooling. These attributes do not directly determine persistence but collectively shape the intellectual and social disposition the student brings to the institution (Tinto, 1975).

2. **Initial Goals and Commitments:** Pre-entry attributes feed into two parallel commitment structures. *Intentions and goal commitments* reflect the student's desire to earn a degree, while *goal and institutional commitments* reflect the desire to graduate from a specific institution. Alongside these sits a third force—*external commitments*—representing obligations outside the institution, such as

family or employment, which compete with academic integration from the outset (Tinto, 1975, 1993).

3. **Institutional Experiences:** Once enrolled, students interact simultaneously with two systems. The *Academic System* operates through both formal channels—academic performance—and informal channels—faculty and staff interactions. The *Social System* operates through formal extracurricular activities and informal peer group interactions. Both systems are embedded within and influenced by the broader *External Community* (Tinto, 1993).

4. **Integration:** The quality of institutional experiences determines the degree of integration achieved along two dimensions. *Academic integration* results from engagement with the Academic System, while *social integration* results from engagement with the Social System. These two dimensions are interdependent—weakness in either increases the probability of departure (Nicoletti, 2019; Tinto, 1975).

5. **Re-evaluated Goals and Commitments:** Integration feeds back into a revised set of commitments—updated intentions, goal and institutional commitments, and external commitments—which are now reconfigured by the student's actual lived experience within the institution. This re-evaluation stage is what makes the model longitudinal and iterative rather than static: commitments are not fixed at entry but continuously reshaped over time (Nicoletti, 2019; Tinto, 1993).

6. **Departure Decision (Outcome):** The re-evaluated commitments produce the final outcome: a *departure decision*. If the cumulative weight of academic and social integration has reinforced the student's goal and institutional commitment, persistence follows. If integration has been insufficient and external commitments or personal costs outweigh the perceived benefits of continued enrollment, departure follows. Critically, the entire process unfolds across time (t), underscoring that dropout is a process, not a moment (Tinto, 1975).

2.3.1.2 The Two Pillars of Integration

Tinto distinguishes two primary and interdependent dimensions of integration, both of which must be achieved to sustain persistence (Tinto, 1993):

Academic Integration encompasses both structural and normative dimensions. Structural integration is measured by grade performance (GPA) relative to institutional standards, while normative integration reflects intellectual development and alignment with the academic community's values. Grades function not merely as performance records but as institutional signals of belonging within the academic system; a declining GPA is therefore not simply a performance metric but a symptom of deteriorating academic integration (Bento Júnior et al., 2025; Tinto, 1975).

Social Integration reflects the quality of peer-group interactions and faculty relationships. Research on blended and online learning environments has confirmed that when instructors provided timely support, appropriate feedback, and interactive engagement, students were substantially more likely to persist (Ivankova & Stick, 2007). Furthermore, studies measuring LMS engagement have established that completers consistently demonstrate significantly higher levels of participation in learning activities—measured through discussion posts, resource access, and login frequency—than withdrawers, validating these behavioral signals as digital proxies for integration (Morris et al., 2005).

2.3.1.3 Limitations and Critiques

Despite its canonical status, the SIM carries limitations directly relevant to this study's context. Its key constructs—integration, commitment, fit—lack precise operational definitions, creating interpretive challenges when translating the model into computable variables (Nicoletti, 2019). The model was also developed within a Western, residential university context and requires careful contextual recalibration before application in non-Western institutions such as Strathmore University. Additionally, traditional SIM-based research has relied predominantly on retrospective, correlational designs: the model identifies *what* predicts departure but does not generate prospective, real-time

predictions for individual students. This retrospective limitation is the precise gap that machine learning addresses—where the SIM identifies the risk factors, ML determines when those factors are currently activating for a specific student.

2.3.1.4 Operationalization in This Study

This study operationalizes Tinto's model within Strathmore's digital infrastructure by systematically mapping each theoretical construct onto a measurable institutional variable. The mapping is deliberate and bidirectional: every predictor variable selected for the ML models corresponds to a specific SIM construct, ensuring that algorithmic outputs retain theoretical interpretability.

2.3.2 Astin's Input-Environment-Output (I-E-O) Model

Alongside Tinto's SIM, Alexander Astin's Input-Environment-Output (I-E-O) model provides a complementary and practically powerful theoretical lens for this study. Originally developed as a conceptual guide for institutional assessment, the model was empirically consolidated in Astin's landmark study *What Matters in College? Four Critical Years Revisited*, which drew on surveys of more than 20,000 students, 25,000 faculty members, and 200 institutions to examine how academic programmes, faculty, peer groups, and institutional policies shape student development and outcomes (Astin, 1993). The model's philosophical foundation is a *talent development* approach to institutional excellence: quality is not measured by institutional prestige or resource endowment, but by the degree to which the institution succeeds in developing the potential of the students it admits (Astin & Antonio, 2012). This orientation makes the I-E-O model especially relevant to the Strathmore context, where the goal is not to rank students but to identify and support those whose potential is at risk of being lost.

2.3.2.1 Structure of the Model

The I-E-O model organizes the variables relevant to student success into three sequential, interacting components (Astin & Antonio, 2012):

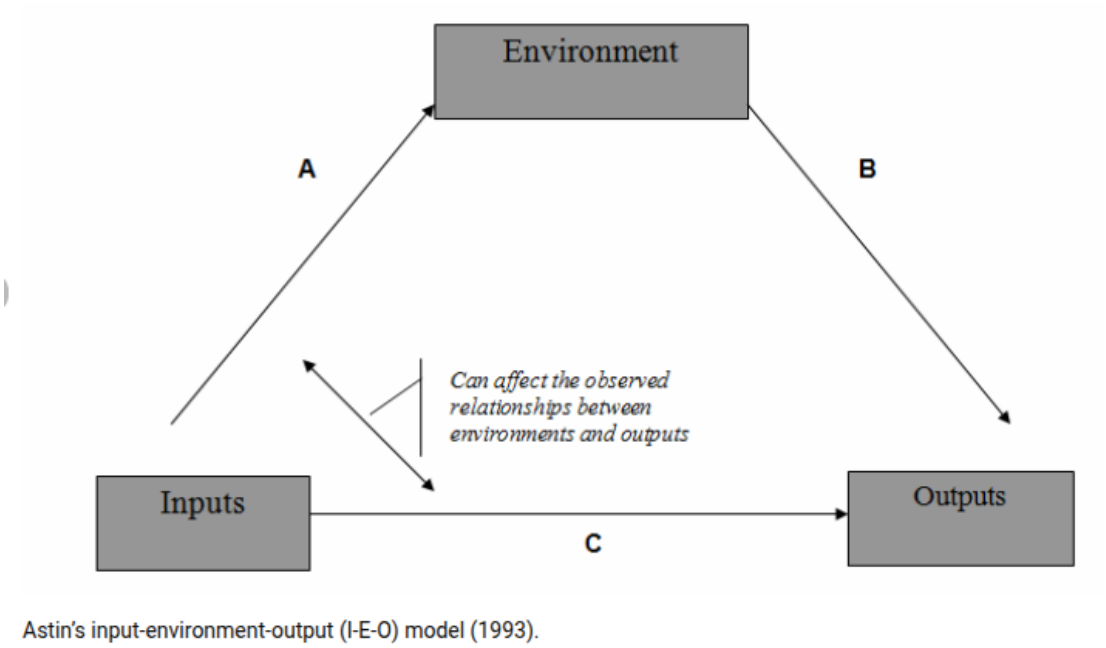

Figure 2.2: Astin's input-environment-output (I-E-O) model (1993).

1. **Inputs** refer to the personal qualities and characteristics students bring with them at the point of entry. This includes demographic attributes (age, sex), family background, prior academic achievement such as secondary school GPA, and initial motivations and goals. Astin's framework treats no single input as deterministic; rather, inputs establish the baseline against which the effects of institutional experiences must be measured (Astin & Antonio, 2012).

2. **Environment** encompasses the full range of institutional experiences, programmes, policies, and interactions to which the student is exposed during enrollment. This includes both tangible elements (classrooms, library resources, residential halls) and intangible elements (teaching pedagogy, administrative policies, peer group composition, and the quality of faculty interaction). Environmental variables are particularly significant in the I-E-O framework because they represent the factors that educators and institutional administrators can directly control to maximize talent development (Astin & Antonio, 2012). Applied research using this framework has confirmed that student-advisor interaction, student-faculty interaction, extracurricular participation, and library utilization are each positively and inde-

pendently associated with first-to-second year retention (Xiao et al., 2020).

3. **Outputs** (or Outcomes) represent the measurable results of the educational programme: academic achievement, retention, degree completion, and the full range of skills, values, and attitudes the institution aims to develop. In the I-E-O framework, outcomes are understood as the product of how environmental conditions transform a student's initial input characteristics. Evaluating outcomes without accounting for inputs is explicitly identified by Astin as an incomplete assessment design, as it fails to capture how much students actually developed relative to their starting point (Astin & Antonio, 2012).

2.3.2.2 Methodological Logic: The Natural Experiment

A defining methodological feature of the I-E-O model is its use of what Astin terms *natural experiments*. Because students cannot be randomly assigned to different institutional environments, the model relies on complex multivariate statistical analyses—such as multiple logistic regression—to control for input differences and isolate the true effect of environmental variables on outcomes (Astin & Antonio, 2012). This approach offers two practical advantages. First, it avoids the artificial conditions of laboratory experiments by studying real institutional variation. Second, it allows researchers to examine the simultaneous effects of many environmental variables, which is precisely the analytical challenge confronted in this study's multi-stream data pipeline. The machine learning models developed in Chapter Three operationalize this same logic: by training on historical student cohort data with known outcomes (2021–2024), the models statistically isolate which combinations of input and environmental variables most strongly predict each risk outcome, mirroring Astin's natural experiment methodology in a computational framework.

2.3.2.3 Complementarity with Tinto's Model

The I-E-O model and Tinto's SIM are theoretically complementary rather than competing. Tinto explains *why* students depart through the mechanism of diminishing integra-

tion; Astin provides the *variable architecture*— the structured categorization of inputs, environmental conditions, and outcomes—that a predictive model must measure (Bento Júnior et al., 2025; Nicoletti, 2019). This complementarity is particularly significant for the three-outcome prediction framework of this study. Dropout risk maps most directly to Tinto's integration failure mechanism. Course failure risk and programme delay risk, however, are better understood through Astin's environmental lens: they reflect whether the institution's academic policies, attendance requirements, and support resources are functioning as effective environmental buffers against disengagement. A student may remain enrolled— not a dropout in Tinto's terms—while still accumulating course failures or programme delays that Astin's output variables would capture.

2.3.2.4 Operationalization in This Study

The I-E-O model maps directly and systematically onto the three institutional data streams integrated in this study's analytics pipeline:

Inputs are operationalized through Academic Management System (AMS) variables: entry qualifications, prior GPA trajectory, school of enrollment, and year of study. These correspond to Astin's input variables and establish the baseline characteristics against which environmental effects are assessed (Astin & Antonio, 2012).

Environmental variables are operationalized through LMS engagement metrics—login frequency, resource access, and assignment submission timeliness—and real-time attendance records. These capture the student's degree of active interaction with the institutional environment. Empirical research applying the I-E-O framework has demonstrated that student-advisor interaction, student-faculty interaction, extracurricular participation, and library utilization are each positively associated with first-to-second year retention (Xiao et al., 2020), directly validating the use of engagement metrics as environmental proxies in this study. Strathmore's specific institutional policies—the 67% attendance threshold and the 2.0 GPA floor— function as measurable environmental constraints that the predictive models are explicitly calibrated to reflect.

Outputs are the three predicted risk outcomes: dropout risk, course failure risk, and programme delay risk. By framing these as distinct output categories, this study extends Astin's output framework beyond the traditional graduation/dropout binary to capture the graduated spectrum of academic difficulty that students actually experience. This extension is methodologically consistent with Astin's own recognition that institutional outcomes are multidimensional and that talent development can be undermined in forms that fall short of full withdrawal (Astin & Antonio, 2012).

2.3.2.5 Limitations and Critiques of the I-E-O Model

Despite its widespread adoption, the I-E-O model carries documented limitations that must be acknowledged. The model was developed within a Western, residential, four-year university context, where full-time on-campus participation is assumed to be the primary channel of environmental engagement (Astin, 1993). This assumption does not transfer without recalibration to commuter, part-time, or non-Western institutional settings (Karp et al., 2008).

A further limitation is the model's linear input-to-environment-to-output structure, which oversimplifies the recursive nature of student development. In practice, outputs such as academic performance continuously reshape subsequent environmental engagement, creating feedback loops that a static framework cannot capture (Astin & Antonio, 2012). These limitations do not invalidate the I-E-O model as a theoretical anchor for this study; rather, they motivate the adoption of a longitudinal machine learning architecture that continuously re-evaluates risk as new engagement and performance data arrives, compensating for the model's structural simplifications in a real institutional deployment context.

2.4 Machine Learning in Educational Data Mining (EDM)

2.4.1 The Paradigm Shift: From Lag Indicators to Predictive Signals

Educational Data Mining (EDM) has undergone a profound methodological transformation over the past decade. Traditional approaches relied on 'lag indicators'—mid-semester grades, end-of-term transcripts, and formal withdrawal notices—which, by the time they are generated, often arrive too late for meaningful academic intervention (Ahmed, 2024). The emergence of machine learning has fundamentally altered this temporal dynamic.

Research by Ahmed (2024) and Gathani et al. (2026) demonstrates that modern ML algorithms can identify at-risk students with prediction accuracies ranging from 88% to 92%, often weeks or months before academic failure becomes irreversible. This temporal advance is the core value proposition of EWS architecture: it converts institutional data from a historical record into a forward-looking instrument.

2.4.2 Model Performance Metrics and Evaluation Standards

In the domain of student risk prediction, accuracy alone is an insufficient evaluation criterion. The asymmetric costs of misclassification—where failing to identify a genuinely at-risk student carries far greater consequences than a false alarm—necessitate a more nuanced suite of evaluation metrics:

- Recall (Sensitivity): The proportion of genuinely at-risk students correctly identified by the model. High recall is the primary objective in EWS contexts, as missed detections represent failed interventions.
- Precision: The proportion of students flagged as at-risk who are genuinely at-risk. Precision governs the credibility of the system; excessively low precision generates alert fatigue among advisors.

- F1-Score: The harmonic mean of Precision and Recall, providing a single balanced metric that is particularly informative when class distributions are imbalanced—a common characteristic of student datasets where at-risk students are a minority.
- ROC-AUC (Receiver Operating Characteristic – Area Under Curve): A threshold-independent measure of the model's ability to discriminate between successful and at-risk students across all possible classification cutoffs. A value approaching 1.0 indicates near-perfect discrimination.

These metrics collectively ensure that model evaluation reflects real-world operational requirements rather than abstract statistical performance.

2.4.3 Ensemble Learning: The Technical Foundation

To achieve the predictive stability required for institutional deployment, this research employs Ensemble Learning—a class of machine learning technique that combines the outputs of multiple algorithms to produce predictions that are more accurate and robust than any single model could achieve in isolation.

The three ensemble algorithms selected for this study each offer distinct advantages. Random Forest constructs multiple independent decision trees and aggregates their predictions through majority voting, reducing variance and mitigating overfitting—a particular risk when working with moderately sized institutional datasets of approximately 5,000 students (Breiman, 2001). Gradient Boosting builds trees sequentially, with each tree correcting the errors of its predecessor, producing a powerful predictor that is especially effective on structured, tabular data characteristic of academic records. XGBoost (Extreme Gradient Boosting), pioneered by Chen and Guestrin (2016), extends the gradient boosting framework with regularization terms that prevent overfitting, efficient handling of missing values, and computational optimization that enables scalable deployment. Its proven performance on structured datasets and widespread adoption in applied predictive analytics make it particularly well-suited to this research context.

The selection of ensemble tree-based methods over deep learning architectures is a deliberate methodological choice justified by two factors. First, the dataset size of approximately 5,000 undergraduate students is insufficient to fully exploit the representational capacity of deep neural networks without significant risk of overfitting. Second, and critically, tree-based ensembles yield feature importance rankings that are inherently more interpretable than deep learning's distributed representations, providing a natural foundation for explainability tools.

2.4.4 The Black Box Problem and Explainable AI (XAI)

The adoption of complex ensemble models in institutional settings is frequently impeded by what the literature terms the 'Black Box' problem: the inability of users—advisors, administrators, and students—to understand why a particular prediction was made (Lundberg Lee, 2017). In a university context, this opacity is not merely a technical inconvenience. Advisors who cannot explain why a student has been flagged lack the conversational foundation to initiate a credible, evidence-led intervention. Students who receive unexplained risk labels may feel stigmatized or surveilled rather than supported.

The most widely adopted framework for addressing this problem is SHAP (SHapley Additive exPlanations), developed by Lundberg and Lee (2017) from cooperative game theory's Shapley values. SHAP assigns each feature a contribution value for each individual prediction, indicating both the direction and magnitude of that feature's influence on the model's output. For a student flagged with high dropout risk, SHAP might reveal that declining LMS engagement over the past three weeks is the primary driver, followed by attendance approaching the 67% threshold and a drop in the most recent coursework grade.

In resource-constrained environments such as Kenyan universities, where institutional trust must be earned incrementally and advisor capacity is limited, explainability is not a luxury feature—it is a prerequisite for adoption. By transforming the advisor's interaction from 'this student is at risk' to 'this student is at risk because of these specific,

documented behaviors,' SHAP-driven dashboards facilitate substantive, reason-led conversations that are both more effective and more defensible.

2.4.5 Risk Outcome Categories in the Literature

The existing EDM literature predominantly frames student risk as a binary problem—a student either persists or drops out. This study extends the literature by predicting three distinct risk outcomes simultaneously, reflecting the multi-dimensional nature of academic difficulty:

- Dropout Risk: The probability that a student will discontinue enrollment before completing their degree—the most consequential outcome for both the student and the institution.
- Course Failure Risk: The likelihood of failing an individual course or becoming examination-ineligible due to attendance falling below the 67% threshold mandated by Strathmore University.
- Program Delay Risk: The probability that a student will exceed the standard programme duration due to academic probation, course repetition, or accumulated failed semesters.

By distinguishing these three outcomes, the platform enables differentiated intervention strategies rather than a single undifferentiated 'at-risk' label, allowing advisors to tailor responses to the specific nature of each student's vulnerability.

2.5 Evidence of Technological Maturity

The most extensively documented implementation of ML-based student success analytics is Georgia State University's GPS Advising system (Georgia State University, 2024). The system tracks over 800 risk factors daily for a student population of approximately 40,000, generating 90,000 targeted advisor interventions each academic year. The outcomes are remarkable in scale: the university achieved a 7.8 percentage point

increase in its six-year graduation rate and virtually eliminated achievement gaps between demographic groups—outcomes that had previously been attributed to structural inequalities resistant to intervention.

The Georgia State case establishes two critical propositions for this research. First, it demonstrates that the technology is mature, proven, and deployable at institutional scale. Second, and more importantly, it illustrates the mechanism through which data-driven intervention closes equity gaps: by systematically identifying students who would otherwise be overlooked by advisor intuition, the system ensures that support is distributed on the basis of need rather than visibility. This principle is directly relevant to the Strathmore context, where advisor capacity is finite and unevenly distributed across a population of approximately 5,000 undergraduate students.

2.6 Predictive Analytics in the African Higher Education Context

2.6.1 Continental Policy Framework

The African Union's Continental Education Strategy for Africa (CESA 2016–2025) formally identifies student retention as a continental priority, calling for systemic improvements in the quality and inclusivity of higher education across member states. Despite this strategic commitment, the operational implementation of retention technology remains nascent across most of the continent. The dominant institutional practice continues to rely on end-of-semester transcripts as the primary diagnostic tool—a paradigm that, by definition, identifies failure only after it has occurred.

This gap between continental policy aspiration and institutional practice defines the landscape in which this study operates, providing both a mandate and an opportunity for evidence-based technology adoption.

2.6.2 Regional Case Studies

A small but growing body of African pilot studies demonstrates that ML can extract actionable risk signals from institutional datasets, even under conditions of data noise and resource constraint. In Morocco, Bouihi et al. (2024) conducted a dropout prediction study at Mohammed V University utilizing L2-regularized logistic regression—a technique that applies a mathematical penalty to prevent the model from overfitting sparse institutional data. The study achieved an accuracy rate of 81.66%, demonstrating the efficacy of regularized models in the African institutional context and establishing a regional benchmark against which more sophisticated ensemble methods can be assessed.

In South Africa, Taodzera (2016) analyzed a cohort of 1,366 engineering students at the University of Johannesburg using Decision Tree algorithms, achieving an accuracy rate of 65.86%. Despite this more modest performance, the study produced a theoretically significant finding: prior performance in physics and mathematics emerged as the primary predictors of engineering student success, suggesting that discipline-specific feature engineering is essential for contextually valid risk prediction. This insight informs the feature selection strategy of the current study.

These regional cases demonstrate that while African institutional datasets may present greater noise than their Western counterparts—due to data quality issues, inconsistent record-keeping, and structural gaps—machine learning can nonetheless extract patterns of sufficient fidelity to support targeted intervention.

2.7 The Kenyan Higher Education Landscape

2.7.1 Structural and Economic Pressures

Within East Africa, Kenya's higher education sector operates under acute financial stress that directly amplifies the consequences of student attrition. Government capitation currently covers only 57% of students against an 80% target, while the accu-

mulated unpaid obligations of public universities have reached Ksh 62 billion (Daily Nation, 2024; KIPPRA, 2024). In this resource-constrained environment, each student who withdraws before completing their degree represents a compounded loss: for the student who exits with debt but without a credential, and for the institution that absorbs the fixed costs of that student's enrollment without recovering the full tuition investment.

This structural context transforms student retention from a pedagogical aspiration into an economic imperative. The status quo of 'universal mentoring'—in which advisor attention is distributed broadly across the student population without systematic risk stratification—is demonstrably unsustainable in an environment of resource scarcity. Evidence-based allocation, directed by automated risk prediction, offers a more efficient and equitable alternative.

2.7.2 National Policy Mandate

The Republic of Kenya Ministry of Education's Sessional Paper No. 1 of 2024 on Reforming Education and Training for Sustainable Development explicitly identifies 'inaccurate, unreliable data and inadequate utilisation of technology' as a core bottleneck impeding the quality and efficiency of the national education system (Republic of Kenya Ministry of Education, 2024). This policy declaration establishes a formal mandate for institutional investment in data-driven student support systems, providing a regulatory context that supports the current research's implementation and replication agenda.

2.7.3 Foundational Kenyan Study: Machakos University College

A foundational study in the Kenyan EDM literature was conducted by Ngemu et al. (2015) at Machakos University College, utilizing the WEKA machine learning toolkit to evaluate multiple classification algorithms on historical institutional data. The study assessed Naïve Bayes, Support Vector Machines (SVM), and the J48 Decision Tree

algorithm, identifying four variables as the primary predictors of student dropout risk:

- Financial stability, specifically measured through fee payment status and timeliness.
- Entry age, reflecting the relationship between non-traditional enrollment age and retention challenges.
- Health status, capturing the impact of chronic illness or disability on academic continuity.
- Parental occupation, serving as a proxy for socioeconomic background and domestic support structures.

While the Machakos study demonstrates that even basic machine learning tools applied to existing institutional records can yield actionable risk signals, it also highlights important limitations relative to the current research. The study relied on a limited set of variables predominantly capturing socioeconomic background rather than real-time behavioral engagement, and it employed single-algorithm approaches rather than the ensemble methods that have since been established as superior in terms of predictive stability. This study extends the Kenyan literature by incorporating LMS engagement metrics as behavioral lead indicators, deploying ensemble learning for enhanced accuracy, and producing SHAP-based explanations that are absent from prior Kenyan research.

2.8 Data Integration and Institutional System Architecture

2.8.1 Siloed Administrative Repositories

A persistent structural challenge in the deployment of institutional EWS is the fragmentation of student data across isolated administrative systems. In most universities, including Strathmore, student information is distributed across at least three distinct plat-

forms: the Academic Management System (AMS), which maintains official records of enrollment, academic history, course grades, and demographic information; the Learning Management System (LMS), which captures behavioral engagement data such as login frequency, resource access, and assignment submission patterns; and attendance tracking infrastructure, which records real-time class participation.

These systems were designed as independent administrative tools rather than integrated analytics infrastructure, and they typically lack the interoperability required for unified risk computation. The consequence is a 'fragmentation gap': institutions possess the raw data necessary for sophisticated risk prediction but lack the pipeline architecture to harmonize and analyze it in real time (Republic of Kenya Ministry of Education, 2024).

2.8.2 Engagement Metrics as Lead Indicators

A critical contribution of the EDM literature has been the identification of LMS engagement metrics as 'lead indicators'—behavioral signals that precede academic decline by days or weeks and therefore provide an earlier intervention window than grade-based measures alone. Research consistently demonstrates that declining login frequency, missed assignment deadlines, and reduced resource access correlate strongly with subsequent academic failure, often manifesting two to four weeks before a course grade reflects the underlying disengagement (Ahmed, 2024; Gathani et al., 2026).

For Strathmore University, where attendance records are maintained in real time and LMS activity is comprehensively logged, these engagement metrics represent an underutilized predictive resource. By integrating them with AMS academic records through a unified analytics pipeline, this study unlocks their lead-indicator value for the first time within the institution's data environment.

2.9 Summary and Identification of Knowledge Gaps

This literature review has traced the intellectual lineage and operational context of this study across four interconnected domains. Theoretically, Tinto's Student Inte-

gration Model provides the conceptual architecture through which student risk is understood: as a function of diminishing academic and social integration that can be detected through behavioral and performance data before withdrawal occurs. Technologically, the EDM literature demonstrates that ensemble machine learning algorithms—particularly Random Forest, Gradient Boosting, and XGBoost—can achieve prediction accuracies of 88–92% on institutional datasets, and that SHAP-based explainability transforms opaque risk scores into advisor-actionable intelligence. Regionally, studies in Morocco, South Africa, and Kenya confirm that ML can extract meaningful risk signals from African institutional data, though existing studies are constrained by limited algorithm sophistication, narrow variable sets, and the absence of real-time integration. Institutionally, Strathmore University possesses the data infrastructure necessary for a sophisticated EWS but currently operates those systems as isolated administrative silos within a reactive support paradigm.

This review identifies a specific and critical gap in the existing literature: no prior study has developed an integrated, real-time risk stratification system that simultaneously satisfies four conditions: (1) harmonizing multiple institutional data streams—AMS, LMS, and attendance records—into a unified analytics pipeline; (2) predicting three distinct and clinically relevant risk outcomes (dropout risk, course failure risk, and program delay risk) rather than a single binary outcome; (3) calibrating predictive models to the specific academic policy thresholds of a Kenyan institution, including the 67% attendance requirement and the 2.0 GPA floor that characterize Strathmore's academic standards; and (4) producing SHAP-driven, role-differentiated explanations that enable non-technical academic advisors to conduct evidence-led, credible interventions.

This study is designed to fill precisely that gap. Chapter Three presents the research methodology through which these four objectives are operationalized, detailing the data pipeline architecture, feature engineering strategy, model development and validation approach, and the design of the role-based dashboard interface.

2.10 Logical Flow of LR

Introduction

Global Perspectives on Student Attrition

Theoretical Frameworks for Student Retention

Machine Learning in Educational Data Mining

Existing Predictive Analytics for at-risk students

2.10.0.1 Georgia State University

2.10.0.2 Moroko

2.10.0.3 South Africa: Joanesbourg

2.10.0.4 South Africa:Cap Town

2.10.0.5 Machakose case

Summary and Knowledge Gaps

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